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A CAPSTONE PROJECT REPORT

ON

“ATM–BANK SERVER COMMUNICATION SIMULATION ”

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

COMPUTER NETWORKS

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “ATM–Bank Server Communication Simulation ” has been carried out by Nibraz M (1965260001) under the supervision of Dr.K.Jaisharma and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, Nibraz M of the Electronics and Communication Engineering Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Computer Networks project work entitled “ATM–Bank Server Communication Simulation ” is my individual academic work. The study is based on a simulated ATM-to-bank communication environment and is intended only for educational purposes. No real banking account, customer record, PIN, or financial transaction is used. The work has been prepared in accordance with academic integrity and the technical observations are based on the stated laboratory workflow and Wireshark analysis.

Place: Chennai

Date: September 2026

Signature of the Student:

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INDIVIDUAL CONTRIBUTION STATEMENT

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution
Nibraz M / 1965260001	ATM–bank communication simulation	Network design, TCP client/server simulation	Connectivity, transaction and packet-capture testing	Chapters 1–6, References & Appendices	100%

ABSTRACT

Modern ATMs are networked endpoints that exchange transaction information with bank-side systems. This report models that interaction using a small software-based network and studies the communication from both a functional and security perspective. Three ATM endpoints are represented as clients, a router provides the network path, and a central bank server acts as the transaction-processing endpoint.

The practical study is divided into three connected views: network reachability, TCP request–response communication, and packet-level inspection. Cisco Packet Tracer can be used to represent the logical topology, while Python socket programming provides a simple client–server model. Wireshark is then used to observe the transport traffic and understand what can be learned from captured packets.

A major engineering observation is that successful TCP delivery should not be confused with confidentiality. TCP provides reliable transport, but encryption must be supplied by an additional security mechanism such as TLS. The experiment therefore links core Computer Networks concepts with a practical security lesson.

Keywords: ATM; Bank Server; Computer Networks; TCP; Wireshark; Packet Analysis; TLS

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbreviation	Description	Unit / Meaning
ATM	Automated Teller Machine	—
IP	Internet Protocol	—
TCP	Transmission Control Protocol	—
LAN	Local Area Network	—
TLS	Transport Layer Security	—
GUI	Graphical User Interface	—
ACK	Acknowledgement	TCP flag
SYN	Synchronize	TCP flag
FIN	Finish	TCP flag

1. PROBLEM DEFINITION

1.1 Background

An ATM is not an isolated electronic device. A transaction request must travel through a communication network to a server that validates and processes the request before sending a response. The same basic pattern appears in many networked applications: a client generates a request, the network transports it, and a server returns the result.

For a classroom experiment, the banking environment can be simplified without using any real financial information. The simplified model is sufficient to demonstrate IP addressing, routing, TCP connections, application messages, packet capture, and security analysis.

1.2 Problem Statement

Design a repeatable academic model of ATM-to-bank communication and inspect the resulting network traffic to determine how the communication behaves at the packet level. The experiment must demonstrate both successful communication and the security limitation of an unencrypted TCP application channel.

1.3 Aim

To design and analyse a simplified ATM-to-bank server communication system using Computer Networks concepts, with Wireshark-based packet inspection of the TCP traffic.

1.4 Objectives

- Create a logical network containing ATM clients, a router and a bank server.
- Assign suitable IP addresses and verify end-to-end reachability.
- Establish TCP communication between an ATM client and the server.
- Demonstrate balance enquiry and withdrawal request/response scenarios.
- Capture the communication using Wireshark.
- Apply practical display filters to isolate relevant packets.
- Interpret TCP fields such as source/destination IP, ports, flags and sequence information.
- Identify the confidentiality risk associated with plaintext application data.
- Suggest TLS-based protection as the security improvement.

1.5 Scope

Included	Not Included
Simulated ATM terminals	Physical ATM hardware
Router-based IP communication	Real banking infrastructure
TCP client/server exchange	Live customer accounts
Balance and withdrawal logic	Real monetary transactions
Wireshark packet inspection	Production security architecture
Academic test data	Actual credentials or PINs

2. NETWORK DESIGN AND TECHNOLOGY BASIS

2.1 Logical Architecture

The proposed communication path is arranged as multiple ATM clients connected to a network intermediary and then to a central server. The router represents the forwarding function between network segments. The bank server receives transaction requests and returns application responses.

Component	Role	Main concept
ATM-1	Transaction client	TCP client
ATM-2	Transaction client	TCP client
ATM-3	Transaction client	TCP client
Router	Forward packets	IP routing
Bank Server	Process requests	TCP server
Wireshark	Observe traffic	Packet analysis

2.2 Suggested Addressing Plan

Node	Example IP	Purpose
ATM-1	192.168.10.11	Client endpoint
ATM-2	192.168.10.12	Client endpoint
ATM-3	192.168.10.13	Client endpoint
Router LAN	192.168.10.1	Default gateway
Bank Server	192.168.20.10	Server endpoint
Router Server-side	192.168.20.1	Gateway

The addresses above are example values for a laboratory configuration. They may be replaced by the addresses actually assigned in the student's Packet Tracer or local test environment.

2.3 Technologies Used

Technology	Purpose in the study
IPv4	Logical addressing and packet delivery
TCP	Reliable connection-oriented transport
Python sockets	Simple client/server message exchange
Cisco Packet Tracer	Logical network construction and connectivity testing
Wireshark	Packet capture, filtering and protocol inspection
TLS	Recommended confidentiality mechanism

2.4 Why TCP?

The transaction model requires dependable delivery and an ordered exchange of messages. TCP establishes a connection before application data is exchanged and provides retransmission, sequencing and acknowledgement mechanisms. However, these properties solve reliability problems; they do not automatically encrypt the application payload.

2.5 Why Wireshark?

Wireshark makes network activity visible at packet level. It can display protocol layers, endpoint addresses, port numbers, TCP flags and application information when that information is available. This makes it suitable for demonstrating the difference between communication that works and communication that is also confidential.

3. IMPLEMENTATION WORKFLOW

3.1 Network Preparation

First create the logical topology, connect the endpoints, assign addresses and configure the router interfaces. Each ATM should have the correct gateway. The server must be reachable from the client network.

Step	Activity	Verification
01	Place ATM clients and bank server	All devices visible
02	Connect devices to the router/switching path	Links active
03	Configure IP addresses	Address table verified
04	Configure gateway/routing	Routes available
05	Test with ping	Replies received

3.2 Transaction Model

The ATM acts as the client. It establishes a TCP connection to the bank server and sends a small application message representing a transaction. The server reads the request, performs the corresponding simulated operation, and sends a response.

Transaction	Client request	Server response
Balance enquiry	BALANCE	Current simulated balance
Withdrawal	WITHDRAW <amount>	APPROVED / DENIED
Invalid request	Unsupported command	Error response

3.3 Verification Sequence

- Confirm physical/logical connectivity in the simulator.
- Confirm IP configuration and routing.
- Ping the bank server from the ATM side.
- Start the TCP server application.
- Start the ATM client and initiate a test request.
- Verify that a response is received.
- Start Wireshark before repeating the transaction.
- Stop capture after the request/response exchange and save the capture for analysis.

3.4 Functional Test Matrix

Test	Expected result	Status to record
ATM → Server ping	Reachability confirmed	Pass / Fail
TCP connection	Connection established	Pass / Fail
Balance enquiry	Balance returned	Pass / Fail
Withdrawal within limit	Approval returned	Pass / Fail
Withdrawal beyond limit	Denial returned	Pass / Fail
Wireshark capture	TCP packets visible	Pass / Fail

4. WIRESHARK PACKET-ANALYSIS EXPERIMENT

This experiment examines the same ATM-to-bank communication from the perspective of captured packets. The objective is not merely to collect packets, but to identify the protocol behaviour and understand what information may be exposed when the application channel is not encrypted.

4.1 Experiment Aim

To capture ATM-to-bank TCP traffic in Wireshark, isolate the relevant packets using display filters, inspect the TCP exchange, and evaluate the security implications of plaintext application communication.

4.2 Requirements

- Computer with Wireshark installed.
- A working test client and server or an equivalent local TCP communication setup.
- The ATM/server IP addresses and server port number.
- A controlled test message that contains no real personal or financial information.

4.3 Capture Procedure

Step	Procedure
1	Open Wireshark and identify the active network interface.
2	Begin packet capture before generating the transaction.
3	Start the bank-server application.
4	Generate one balance enquiry and one withdrawal request from the ATM client.
5	Allow the server to return the responses.
6	Stop the capture and save the packet trace.
7	Apply display filters to isolate the relevant TCP conversation.
8	Select a packet and inspect Ethernet/IP/TCP/application details.
9	Use Follow → TCP Stream when appropriate to reconstruct the test conversation.
10	Record the observation without using real credentials or sensitive data.

4.4 Useful Wireshark Display Filters

Purpose	Display filter
All TCP traffic	tcp
Traffic involving the server IP	ip.addr == 192.168.20.10
Specific server port	tcp.port == 5000
Client-to-server direction	ip.dst == 192.168.20.10 && tcp
TCP connection setup	tcp.flags.syn == 1
TCP connection termination	tcp.flags.fin == 1
Packets containing a specific test term	tcp contains "BALANCE"

Replace the example IP address and port with the values used in the actual experiment. The last filter should only be used when the test application sends the selected text as visible TCP payload.

4.5 TCP Connection Behaviour

A normal TCP connection begins with a three-way handshake. The client sends a SYN, the server responds with SYN-ACK, and the client completes the setup with ACK. Application data is exchanged only after the

connection is established. The exact packet numbers depend on the capture and should therefore be recorded from the student's own trace.

TCP stage	Typical flag pattern	Meaning
Connection request	SYN	Client asks to establish a TCP connection
Connection response	SYN, ACK	Server accepts and acknowledges
Connection completion	ACK	Client confirms establishment
Data transfer	ACK + payload	Application information is carried
Closing	FIN / ACK	Connection is terminated

4.6 Packet-Observation Sheet

Use the following table while examining the actual capture. The fields are intentionally left for direct entry from Wireshark rather than inventing packet numbers.

Item	Value to record from Wireshark
Client IP	_____
Server IP	_____
Client port	_____
Server port	_____
SYN packet no.	_____
SYN-ACK packet no.	_____
First application-data packet no.	_____
Response packet no.	_____
FIN/connection-close packet no.	_____

4.7 Interpreting a Captured Transaction

For a balance enquiry, the analyst should first identify the TCP stream associated with the ATM client and server. The IP addresses establish the endpoints, while the TCP ports identify the application connection. The packet details can then be expanded layer by layer to inspect the transport information and any visible payload.

If the application protocol sends readable test text over an unencrypted TCP connection, a packet capture may expose that text. This is the key security observation of the experiment. A working connection therefore does not imply that the information travelling through it is private.

4.8 Security Finding

Observation	Interpretation	Risk
TCP packets are visible	The communication path is observable	Traffic metadata can be analysed
Ports and IPs are visible	Endpoints can be identified	Network information is exposed
Plain application data may be readable	No confidentiality at application layer	Sensitive information could be disclosed
TLS is added	Payload becomes cryptographically protected	Confidentiality is improved

4.9 Recommended Secure Design

The next version of the system should use TLS between the ATM application and the bank server. TLS protects application data while retaining TCP as the underlying reliable transport. In a real banking system, additional controls such as strong authentication, certificate validation, transaction logging, access control and secure key management would also be required.

5. RESULTS AND ENGINEERING DISCUSSION

5.1 Overall Result

The proposed model provides a single workflow that connects network configuration, client–server communication and packet analysis. The functional portion verifies that a transaction request can be transmitted and answered. The Wireshark portion demonstrates how the same exchange can be viewed from the network observer's perspective.

5.2 Result Summary

Area	Expected outcome	Evaluation
Network connectivity	ATM reaches bank server	To be verified in lab
TCP communication	Connection and request/response succeed	To be verified in lab
Transaction logic	Balance/withdrawal response generated	To be verified in lab
Packet capture	Relevant TCP packets visible	To be verified in Wireshark
Security analysis	Plaintext exposure risk understood	Concept demonstrated

5.3 Strengths

- Combines several Computer Networks concepts in one practical model.
- Uses a repeatable software-based environment.
- Provides a clear client–server communication pattern.
- Introduces packet-level analysis instead of relying only on application output.
- Shows why reliability and confidentiality are different engineering requirements.

5.4 Limitations

- The system is an educational simulation and not a banking product.
- The sample topology does not represent the complexity of a real bank network.
- No real customer database or financial account is involved.
- Authentication and authorization are simplified.
- TLS is proposed as the security enhancement rather than treated as a complete production security architecture.

5.5 Risk Assessment

Risk	Cause	Mitigation
Data disclosure	Plaintext application traffic	Use TLS
Unauthorized access	Weak/simplified authentication	Strong authentication and access control
Packet manipulation	Unprotected channel	Authenticated encryption / TLS
Incorrect routing	Address or gateway errors	Configuration checks and testing
False test results	Capture started too late	Start Wireshark before generating traffic

6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This individual Computer Networks study demonstrates the complete path of a simplified ATM transaction: an endpoint creates a request, the network transports it using IP and TCP, the bank server processes it, and a response returns to the client. Wireshark adds a packet-level view that reveals how the communication is represented on the wire.

The most important technical conclusion is that TCP reliability is not the same as communication security. If an application sends sensitive information without encryption, packet capture can expose information that the application intended to keep private. TLS should therefore be placed above TCP whenever confidentiality and integrity are required.

6.2 Individual Learning Outcomes

- Understanding of client–server communication.
- Practical awareness of IP addressing and routing.
- Understanding of TCP connection establishment and termination.
- Ability to start, filter and inspect a Wireshark capture.
- Ability to relate packet fields to application behaviour.
- Awareness of confidentiality risks in unencrypted protocols.
- Improved technical documentation and experiment reporting skills.

6.3 Future Enhancement Plan

Priority	Enhancement	Benefit
High	Add TLS to the socket connection	Confidential and authenticated transport
High	Introduce a database	Persistent account/transaction model
High	Add secure authentication	Better access control
Medium	Transaction logging	Traceability and auditing
Medium	Fund transfer feature	Broader transaction coverage
Low	Improve GUI	Better demonstration experience

6.4 Final Statement

The project demonstrates that network engineering should be evaluated from both the functional and security perspectives. A transaction that succeeds at the application level may still have a serious confidentiality weakness at the communication level. Combining simulation, socket communication and Wireshark analysis provides a useful foundation for understanding that distinction.

REFERENCES

1. Computer Networks course notes and laboratory materials.
2. Cisco Packet Tracer networking and simulation resources.
3. Python standard-library socket programming concepts.
4. Wireshark User's Guide and display-filter documentation.
5. General Transport Layer Security (TLS) concepts for secure network communication.

APPENDIX A — WIRESHARK QUICK CHECKLIST

- Start Wireshark before generating the test transaction.
- Select the correct active interface.
- Identify the client and server IP addresses.
- Identify the TCP server port.
- Use `tcp or tcp.port == <port>` to narrow the capture.
- Locate SYN, SYN-ACK and ACK packets.
- Locate the first application-data packet.
- Inspect the TCP payload only when the test data is intentionally non-sensitive.
- Use Follow TCP Stream for conversation-level analysis.
- Record actual packet numbers from the capture rather than copying sample values.

APPENDIX B — VIVA / ORAL-EXAM POINTS

Question	Short answer
Why is TCP used?	It provides reliable, ordered, connection-oriented delivery.
Does TCP encrypt data?	No. Encryption requires an additional mechanism such as TLS.
What does Wireshark do?	Captures and analyses network packets.
What is a display filter?	A filter used to show only packets matching a condition.
What is the TCP three-way handshake?	SYN, SYN-ACK and ACK.
Why start capture before the transaction?	To avoid missing the connection setup and early packets.
Why avoid real banking data?	The experiment is educational and should not expose sensitive information.
What improves confidentiality?	TLS/secure application-layer encryption.

APPENDIX C — STUDENT INFORMATION

Field	Details
Student Name	Nibraz M
Register Number	1965260001
Course	Computer Networks
Submission Type	Individual Report
Primary Practical Tool	Wireshark
Network / Simulation Tool	Cisco Packet Tracer
Programming Technology	Python TCP sockets
Academic Year	2026

Prepared as an individual academic report. Replace observation-sheet blanks with values from the actual laboratory capture before submission.